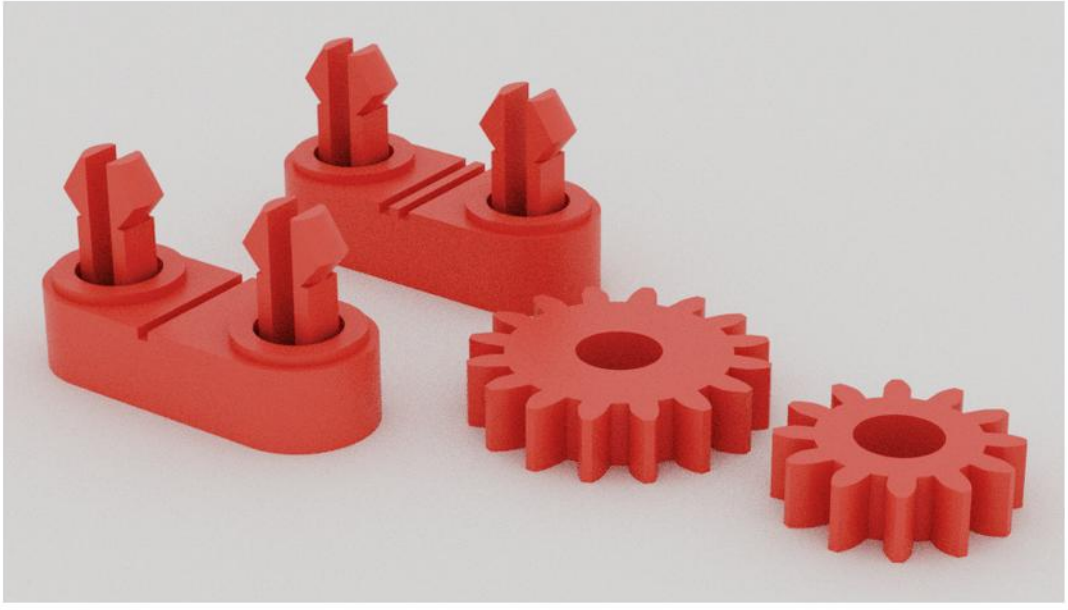
Gear heart test: printing

One plate, about 37 minutes. Before the heart is built around these parts.

1 Print it flat, as it comes



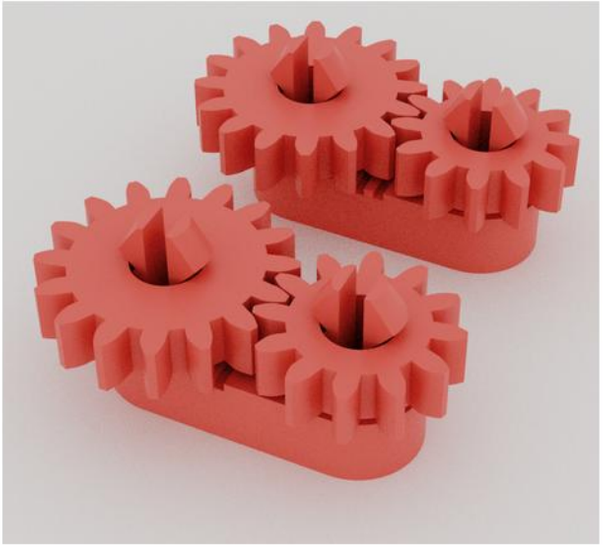
- **gear_heart_test.3mf, any one colour. 0.2 mm layers, supports off.**
- Two bars with snap posts, and one pair of the heart's gears (16 and 12 teeth).
- The bars differ only in how far apart their posts are. 1 notch = tighter, 2 = looser.
- About 37 min and 9.1 g in Bambu Studio.

Design-review renders of the real parts, not photos. The colour is only for the pictures.

Gear heart test: what to check

Design-review renders of the real parts, not photos.

1 Press the gears onto the 1-notch bar



- **Line a gear's hole up with a post and press straight down until it clicks.**
- It takes a firm push, about 11 N: a little more than the weight of a 1 kg bag of sugar. The post's two prongs squeeze in, then spring back.
- Turn one gear: the other should turn with it, smoothly, all the way round.

2 Pull them off, press them onto the 2-notch bar

- **Pull each gear straight up, hard: about 14 N. That tests the barb.**
- Do this once only: the prongs are made to go on and stay on.
- Turn them again and compare with the first bar.

Good

- Clicks on, then spins freely.
- Turns its partner smoothly, no catching.
- Doesn't lift off when you turn it.

Tell me

- which bar meshes better (1 or 2 notches),
- if a gear was too hard to press on, or came off too easily,
- and if a prong cracked.

Design-review renders of the real parts, not photos. The colour is only for the pictures.